

Why $\theta = p_0 = 2/9$

Probability and Phase as Two Readings of One Quotient

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Why $\theta = p_0 = 2/9$: Probability and Phase as Two Readings of One Quotient

Abstract

The Koide angle $\theta = 2/9$ (Doc. 292) and the icosahedral projection $p_0 = |\langle v_0 | R_5 v_e \rangle|^2 = 2/9$ (Doc. 293) arise from two independent calculations and land on exactly the same rational number. This document names the reason and draws the complete consequence.

The reason: both quantities are the same quotient $2/3^2$ – numerator = number of non-trivial $\mathbb{Z}_3$ modes, denominator = square of the $\mathbb{Z}_3$ order. θ reads it as the phase of the mass operator, p_0 as the modulus squared of an algebraic projection. This is the Born-rule structure: the same number, once as a modulus and once as a modulus squared of the same matrix element.

The consequence: the single matrix element $A = \langle v_0 | R_5 v_e \rangle$ of the A_5 embedding determines the Koide formula completely and parameter-free. Its modulus supplies the amplitude coefficient $\sqrt{2} = 3|A|$, its modulus squared supplies the phase $\theta = |A|^2 = 2/9$:

$$a_k = 1 + 3|A| \cdot \cos(|A|^2 + \frac{2\pi k}{3}), \quad A = \langle v_0 | R_5 v_e \rangle.$$

No free parameter. No external θ . The mass amplitudes of the charged leptons follow directly from the A_5 geometry.

What remains transcendental: $\cos(2/9)$, since $2/9 \notin \pi\mathbb{Q}$. Exact algebraic mass amplitudes are therefore excluded in principle.

On the Language: Why “Probability”?

The expression $p_0 = |\langle v_0 | R_5 v_e \rangle|^2$ is called a *transition probability* in quantum mechanics – borrowed vocabulary that can mislead. Neither a statistical experiment nor a frequency is meant. p_0 is the *modulus squared of an algebraic projection*: v_e and v_0 are orthonormal eigenmodes of the $\mathbb{Z}_3$ circulant, $R_5 \in A_5$ is an exact rotation matrix. The inner product $\langle v_0 | R_5 v_e \rangle$ follows entirely from the geometry of A_5 – no random element, no frequency.

The actual finding is algebraic: the same rational number $2/3^2$ appears in two algebraic roles – as the modulus squared of a projection and as the position parameter of the mass operator. The language changes; the object is the same.

External parallel [E]. The same quotient $2/3^2$ appears independently in the conformal field theory of $\mathbb{Z}_3$ orbifolds: the two twist fields σ_1, σ_2 have conformal weights $h_k = k(3-k)/(2 \cdot 3^2) = 1/9$, hence $h_1 + h_2 = 2/9$ (Dixon, Friedan, Martinec, Shenker 1987). There too the numerator counts the non-trivial $\mathbb{Z}_3$ sectors and the denominator is $|\mathbb{Z}_3|^2$. This is a *different* mathematical object – a sum of conformal dimensions, not the modulus squared of a projection – and it arises from a different $\mathbb{Z}_3$ action: the central element $\omega \cdot 1$ on $\mathbb{C}^3$, not the cyclic permutation $C_3 \in A_5$ with eigenvalues $(1, \omega, \omega^2)$. The parallel shows only that $2/3^2$ is a $\mathbb{Z}_3$ -structural counting quantity and not an FFGFT-specific number. No identification of the two objects is claimed.

Epistemic markers:

- [B]** algebraically proved (Python check script, exact arithmetic)
- [K]** numerically confirmed
- [S]** structurally plausible, identification open
- [E]** established external mathematics or physics

Symbol	Meaning
$\mathbb{Z}_3$	cyclic group of order 3; triality on T^4
v_0, v_1, v_2	Fourier eigenmodes of the $\mathbb{Z}_3$ circulant on $\mathbb{C}^3$
ω	$e^{2\pi i/3}$, primitive cube root of unity
A_5	icosahedral group; contains C_3 as a subgroup (Doc. 285)
R_5	five-fold rotation by 72° in A_5
θ	Koide phase; $\sqrt{m_k} = M(1 + \sqrt{2} \cos(\theta + 2\pi k/3))$
p_0	$ \langle v_0 R_5 v_e \rangle ^2$, transition probability
Q	Koide invariant $(\sum m_k)/(\sum \sqrt{m_k})^2 = 2/3$

1 Starting Point: Two Witnesses, One Value

Doc. 293 records the finding as follows:

Route	Origin	Value
Koide (phenomenological)	measured lepton masses; no geometry	$\theta = 2/9$
C_3 -in- A_5 (geometric)	five-fold redistribution of the modes; no masses	$p_0 = 2/9$

The two routes know nothing of each other and land on exactly the same rational number. The stress tests of Doc. 293 (200 random axes, angle scan, order series) show that $2/9$

is icosahedron-specific, not a generic value. Doc. 293 concludes that θ is an “invariant of translation” between differential and spectral geometry – and leaves open why precisely $\theta = p_0$ holds, rather than $\theta = \sqrt{p_0}$ or some other function. That gap is closed here. Section 5 goes one step further: the Koide formula is rewritten entirely in matrix elements of R_5 , so that the mass amplitudes a_k follow directly from the A_5 geometry – without free parameter and without external θ .

2 The Common Quotient $2/3^2$

p_0 as a Count in the $\mathbb{Z}_3$ Mode Space

The three lepton modes are the Fourier eigenmodes of the $\mathbb{Z}_3$ circulant on the $\mathbb{C}^3$ fibre (Docs. 282, 288):

$$v_0 = \frac{1}{\sqrt{3}}(1, 1, 1), \quad v_{1,2} = \frac{1}{\sqrt{3}}(1, \omega^{\pm 1}, \omega^{\pm 2}).$$

One mode is trivial (v_0 , symmetric under $\mathbb{Z}_3$), two are non-trivial (v_1, v_2 , Galois-conjugate). The electron mode is one of the two non-trivial ones (which one is decided by the sign of the angle, Doc. 293).

When $v_e = v_1$ is rotated by R_5 , the mode redistributes over all three directions with the weights (Doc. 293, **[B]**):

$$p_0 = \frac{2}{9}, \quad p_2 = \frac{5-3\varphi}{9}, \quad p_1 = \frac{2+3\varphi}{9}, \quad p_0 + p_1 + p_2 = 1.$$

All three denominators are $9 = 3^2$. For p_0 , the numerator is the small integer 2 – the number of non-trivial modes. In words:

$$p_0 = \frac{\# \text{ non-trivial } \mathbb{Z}_3 \text{ modes}}{|\mathbb{Z}_3|^2} = \frac{2}{3^2}.$$

The denominator is the square, not the group order itself, because p_0 is the modulus squared of an amplitude: $|\langle v_0 | R_5 v_e \rangle| = \sqrt{2}/3$, and $(\sqrt{2}/3)^2 = 2/9$. The factor 3 in the amplitude’s denominator is the normalisation of the modes ($1/\sqrt{3}$ per vector, two vectors in the inner product); the factor $\sqrt{2}$ in the numerator comes from the icosahedral geometry (axis angle 37.38° between the three-fold and five-fold axes, Doc. 293).

θ as a Phase in the $\mathbb{Z}_3$ Mode Space

The Koide parametrisation (Doc. 292)

$$\sqrt{m_k} = M(1 + \sqrt{2} \cos(\theta + 2\pi k/3)), \quad k = 0, 1, 2,$$

carries two structural numbers: the amplitude $\sqrt{2}$ and the phase θ . Doc. 353 shows that $\sqrt{2}$ follows from $\mathbb{Z}_3$ equidistribution **[B]**. Doc. 291 shows in which channel θ lives: not in the symmetric modulus functional (where $Q = 2/3$ for every θ), but in the phase channel of the $\mathbb{Z}_3$ holonomy.

The empirical value is $\theta = 2/9$ (Doc. 292: the PDG back-calculated value deviates from $2/9$ by only 0.9σ , τ -limited; the exact decimal depends on the m_τ central value used, the σ assessment does not). Read as a fraction, $\theta = 2/3^2$. Here too the numerator 2 stands for the two non-trivial modes and the denominator 3^2 for the square of the $\mathbb{Z}_3$ dimension.

Why the square? Because θ measures the phase on the *product space* of the modes – not on the group $\mathbb{Z}_3$ (where the natural fraction would be $2/3$, the share of non-trivial elements), but on $\mathbb{C}^3 \otimes \mathbb{C}^3$, the space of mass-operator elements. The Hermitian circulant M_{ij} has $3^2 = 9$ entries, and the phase that pulls M away from the symmetry axis is a fraction of that 9-dimensional space.

The Same Asymmetry, Read Twice

Both quantities count the same thing:

$$\theta = p_0 = \frac{\# \text{ non-trivial } \mathbb{Z}_3 \text{ modes}}{|\mathbb{Z}_3|^2} = \frac{2}{3^2} = \frac{2}{9}. \quad (1)$$

p_0 reads this quotient as a *probability*: how much of the electron mode lands in the symmetric direction under the icosahedral rotation? θ reads it as a *phase*: how far is the mass configuration from full $\mathbb{Z}_3$ symmetry? Both times the asymmetry of the lepton configuration relative to full $\mathbb{Z}_3$ symmetry is measured – once as a share, once as an angle.

3 The Structure of the Born Rule

This is the structure of quantum mechanics. The Born rule **[E]** links a complex amplitude $A = \langle \psi | \phi \rangle$ to a measurable probability:

$$P = |A|^2.$$

The phase of A is not directly measurable; only $|A|^2$ is. Yet the phase carries the interference information – it becomes visible as soon as two amplitudes are superposed.

The Koide structure exhibits the same dual character, in a particularly simple form:

- The amplitude of the icosahedral projection is $|\langle v_0 | R_5 v_e \rangle| = \sqrt{2}/3$ (modulus; not an adjustable parameter, Doc. 293).
- Its square is the probability $p_0 = 2/9$.
- The Koide angle $\theta = 2/9$ is *numerically the same quotient*, read as a phase.

The relation is not $\theta = |A|$ (which would be $\sqrt{2}/3 \approx 0.471$) and not $\theta = \arg A$ (which would be a rotation angle of the projection), but $\theta = |A|^2 = p_0$. That is the core: *the phase of the mass configuration equals the probability of the mode redistribution*, because both realise the quotient (1).

That the same number appears in both roles is not a numerical accident but a property of the $\mathbb{Z}_3$ mode space: the denominator 3^2 is forced in both roles – as a normalisation square in p_0 , as the dimension of the operator space in θ . The numerator 2 is in both roles the number of non-trivial modes. This is the algebraic form of what Doc. 293 describes as an “invariant of translation”: the number belongs to the structure, not to the representation.

4 Numerical Confirmation

The check script `pruef_367_phase_wahrscheinlichkeit.py` (2/python/Dok367_Skripte) confirms **[K]**:

1. $p_0 = |\langle v_0 | R_5 v_1 \rangle|^2 = 2/9$ to 40 digits (mpmath), with the standard embedding $C_3 < A_5$ (axis $(0, 1, \varphi)$).
2. $|\langle v_0 | R_5 v_1 \rangle| = \sqrt{2}/3$ exactly.
3. θ_{emp} from the PDG masses (back-calculated via $\cos \theta$ from $\sqrt{m_\tau}, \sqrt{m_e}, \sqrt{m_\mu}$; $m_\tau = 1776.86 \pm 0.12$ MeV): 0.222270, deviation from $2/9$: $4.8 \cdot 10^{-5}$; the τ -limited tolerance is $5.3 \cdot 10^{-5}$, so the deviation is 0.9σ (as in Doc. 292).
4. $\theta_{\text{emp}} = p_0$ within the τ tolerance (empirically); exactly $= 2/9$ on both sides (geometrically).
5. The alternatives $\theta = \sqrt{p_0}$ (0.471) and $\theta = \arg \langle v_0 | R_5 v_1 \rangle$ do not reproduce the empirical angle.

5 The Direct Translation: From the U Matrix Element to Lepton Masses

This equation establishes the complete connection. It links the geometry of the A_5 embedding directly to the mass amplitudes of the charged leptons, without any intermediate step or free parameter **[B]**:

$$a_k = 1 + 3 |\langle v_0 | R_5 v_e \rangle| \cdot \cos\left(|\langle v_0 | R_5 v_e \rangle|^2 + \frac{2\pi k}{3}\right). \quad (2)$$

1. The amplitude coefficient ($\sqrt{2}$). The modulus of the transition element is $|\langle v_0 | R_5 v_e \rangle| = \sqrt{2/9} = \sqrt{2}/3$. Scaling this modulus by the dimension of the $\mathbb{Z}_3$ symmetry axis (3) yields exactly the Koide factor:

$$3 \cdot |\langle v_0 | R_5 v_e \rangle| = 3 \cdot \frac{\sqrt{2}}{3} = \sqrt{2}.$$

2. The phase shift (θ). By the Born rule, the modulus squared of the same matrix element is the projection probability p_0 :

$$p_0 = |\langle v_0 | R_5 v_e \rangle|^2 = \frac{2}{9}.$$

Since this square enters the cosine directly as the phase ($\theta = p_0 = 2/9$ rad), the Born rule appears as a geometric coupling: the amplitude and the phase of the same matrix element jointly determine the mass structure.

3. The mass amplitudes. For $k \in \{0, 1, 2\}$, the three Koide amplitudes follow without further assumptions **[K]**:

$$\begin{aligned} \text{Tau } (\tau, k = 0) : \quad a_0 &= 1 + \sqrt{2} \cos\left(\frac{2}{9}\right) \approx 2.3794, \\ \text{Electron } (e, k = 1) : \quad a_1 &= 1 + \sqrt{2} \cos\left(\frac{2}{9} + \frac{2\pi}{3}\right) \approx 0.0404, \\ \text{Muon } (\mu, k = 2) : \quad a_2 &= 1 + \sqrt{2} \cos\left(\frac{2}{9} + \frac{4\pi}{3}\right) \approx 0.5802. \end{aligned}$$

The known mass ratios $m_k \propto a_k^2$ follow immediately.

Assessment. The open question is thereby answered:

- The matrix element $\langle v_0 | R_5 v_e \rangle$ controls both the amplitude ($\propto |A|$) and the phase ($\propto |A|^2$).
- No unbound θ is needed as external input – the number $2/9$ generates itself within the function call from the modulus of the matrix element.
- The intrinsic transcendence via the cosine function ($\cos(2/9)$ is transcendental since $2/9 \notin \pi\mathbb{Q}$) **[E]** remains the mathematical interface, which is why the mass ratios are irreducibly transcendental numbers.

6 What Is Explained and What Remains Open

Explained. The agreement $\theta = p_0$ is not a coincidental convergence of two independent calculations but the double realisation of one quotient $2/3^2$ – once as a probability (modulus squared of a normalised amplitude), once as a phase (share of the operator space). The reason it is $|A|^2$ and not $|A|$ lies in the denominator: the mode space is $\mathbb{C}^3$, the operator space is $\mathbb{C}^3 \otimes \mathbb{C}^3$, and the phase sits in the operator space.

Placement. That phase and probability carry the same number is the language in which FFGFT and quantum mechanics share one structure: amplitude, modulus squared, interference. FFGFT gives that structure a geometric address – $2/3^2$ on the $\mathbb{Z}_3$ mode space – and quantum mechanics gives it the measurement prescription. Both say the same thing, in different languages.